

Final

Megan Wien

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Set Up

```
# reading in packages
library(tidyverse)
library(DHARMA)
library(ggeffects)
```

```
# creating new dataset
nests <- read.csv("data/nest_data_final.csv")
```

Problem 1

a.

In part 1 the test they used a logistic regression test.

In part 2 the test used was a Chi square test of independence.

b.

Part 1: American Avocet microhabitat use was not determined by the distance of the microhabitat from the water.

Part 2: The American Avocet, Forster's Tern, and Black-necked Stilts all occupy different habitats within the managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat study areas.

c.

One other variable that could influence the birds' habitat use is how deep the water is in different microhabitats. This is a continuous quantitative variable that would be measured in meters. It could be measured with a meter stick at the different sites. This could effect the American Avocet habitat use because the depth of the water could determine what prey live in the habitat and the resources available to the birds.

Problem 2

a.

```
# creating new object
nests_clean <- nests |>
  # filtering to only include AB, RRB, and BBR
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # Only using the 3 species
  mutate(
    ant_species = factor(
      ant_species,
      levels = c("AB",
                  "RRB",
                  "BBR")
    ),
    # creating new column of scientific names
    scientific_name = case_when(
      # adding in the scientific names for the species
      ant_species == "AB" ~ "C. sjostedti",
      ant_species == "RRB" ~ "C. mimosae",
      ant_species == "BBR" ~ "C. nigriceps"
    ) |>
  # selecting columns of interest
  select(case_control, height_cm,
         scientific_name)
# showing structure of dataframe
str(nests_clean)
```

'data.frame': 270 obs. of 3 variables:

```
$ case_control : int 1 0 0 0 0 1 0 0 0 1 ...
$ height_cm    : num 392 310 513 154 324 ...
$ scientific_name: chr "C. mimosae" "C. mimosae" "C. mimosae" "C. mimosae" ...
```

```
# displaying 10 rows
slice_sample(nests_clean, n = 10)
```

	case_control	height_cm	scientific_name
1	0	210.5	C. mimosae
2	0	143.0	C. nigriceps
3	0	406.0	C. mimosae
4	1	390.0	C. nigriceps
5	0	243.5	C. mimosae
6	0	159.0	C. mimosae
7	0	329.0	C. mimosae
8	0	156.0	C. sjostedti
9	0	199.0	C. mimosae
10	1	407.5	C. mimosae

b.

The 1s show when a tree had a nest in it and the 0 shows where there was an available tree with no nest.

c.

Tree height is a continuous quantitative variable measured in cm. It could affect the probability of nest occurrence because as the tree height increases so does the probability of there being a nest also increases because it becomes more and more removed from predators. I found this in the papers introduction.

Ant species is a categorical variable of either C. sjostedi, C. mimosae, or C. nigriceps. The species of ants might effect the probability of a nest because they might have different ways of defending themselves from the birds that drives away the birds from making a nest. I found this in the end of the introduction.

d.

model number	description	predictors	interaction term?
1	case_control ~ height_cm + ant_species	tree height and ant species	no

model number	description	predictors	interaction term?
2	case_control ~ height_cm * ant_species	tree height and ant species and the interaction	yes

e.

```
# Model 1: no interaction
model1 <- glm(
  case_control ~ height_cm + scientific_name, # setting variables
  data = nests_clean, # assigning dataset
  family = binomial
)

# Model 2: with interaction
model2 <- glm(
  case_control ~ height_cm * scientific_name, # setting variables
  data = nests_clean, # assigning dataset
  family = binomial
)
```

f.

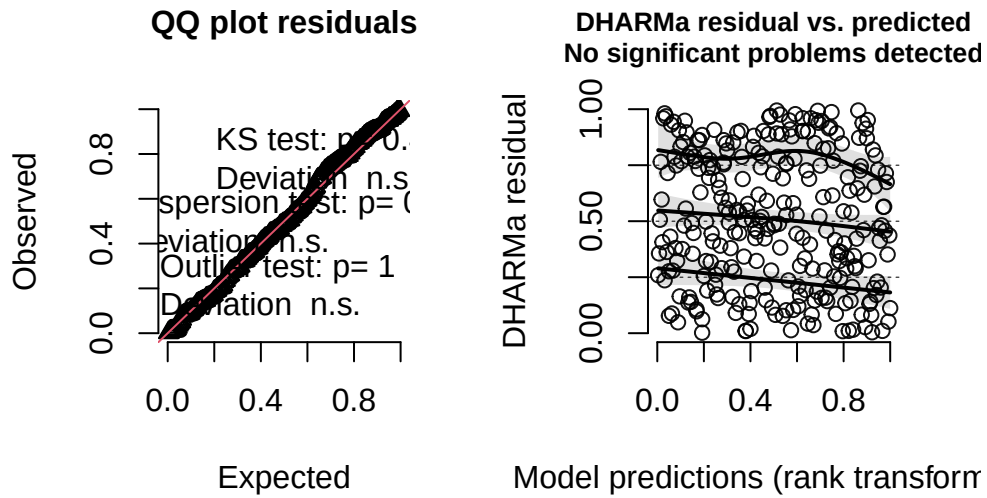
The better model is the interaction model with height times ant species because it had a lower AIC.

g.

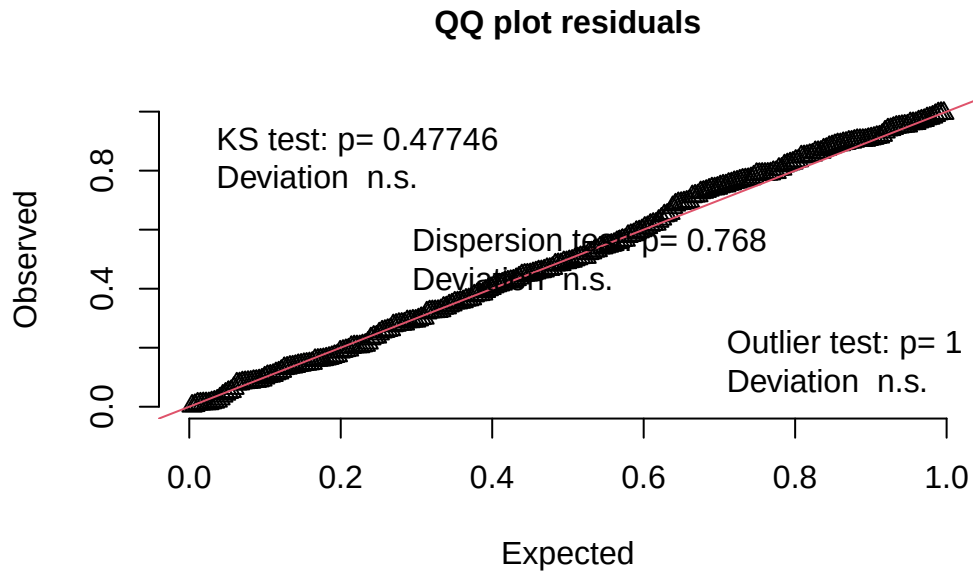
```
# simulate residuals from model2
model2_res <- simulateResiduals(fittedModel = model2)

# plotting the residuals
plot(model2_res)
```

DHARMA residual



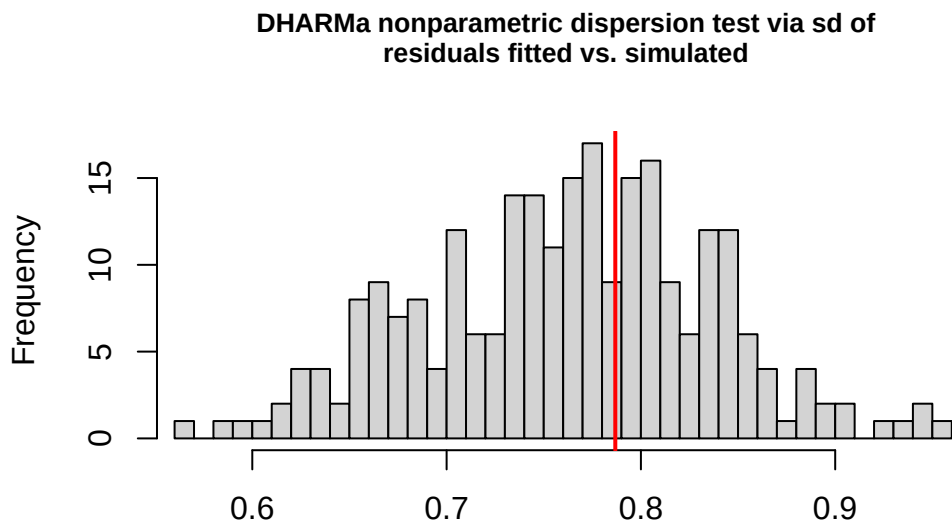
```
# running diagnostic for uniformity
testUniformity(model2_res)
```



Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.051246, p-value = 0.4775  
alternative hypothesis: two-sided
```

```
# running diagnostic for dispersing  
testDispersion(model2_res)
```

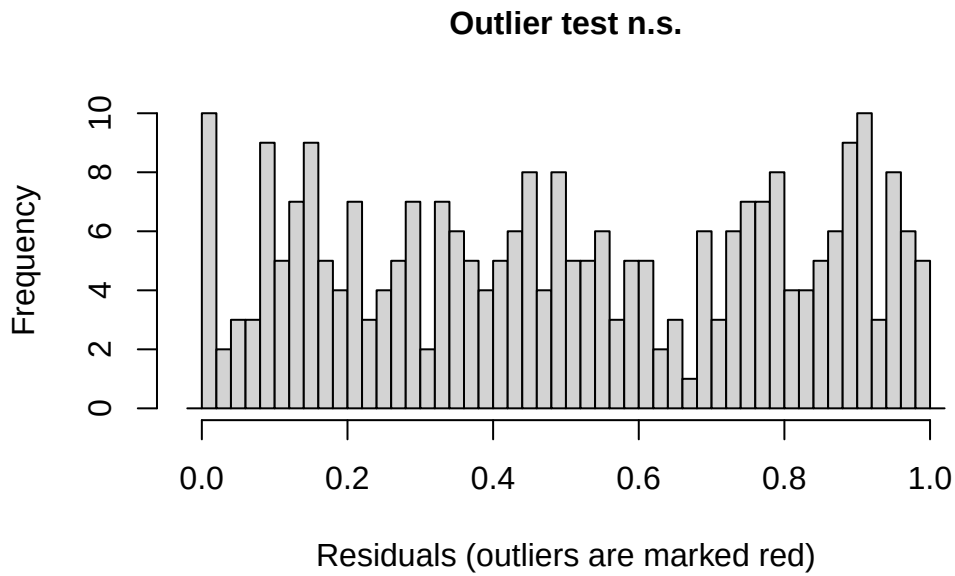


Simulated values, red line = fitted model. p-value (two.sided) = 0.768

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput  
dispersion = 1.0325, p-value = 0.768  
alternative hypothesis: two.sided
```

```
# running diagnostic for outliers  
testOutliers(model2_res)
```



DHARMA bootstrapped outlier test

```
data: model2_res
outliers at both margin(s) = 0, observations = 270, p-value = 1
alternative hypothesis: two.sided
percent confidence interval:
 0.0000000000 0.0019444444
sample estimates:
outlier frequency (expected: 0.000111111111111111 )
0
```

h.

```
# creating predictions object from model 2
predictions <- ggpredict(
  model2,
  terms = c("height_cm", "scientific_name")
)
```

Some of the focal terms are of type `character`. This may lead to

unexpected results. It is recommended to convert these variables to factors before fitting the model.

The following variables are of type character: `scientific\_name`

Data were 'prettified'. Consider using `terms="height\_cm [all]"` to get smooth plots.

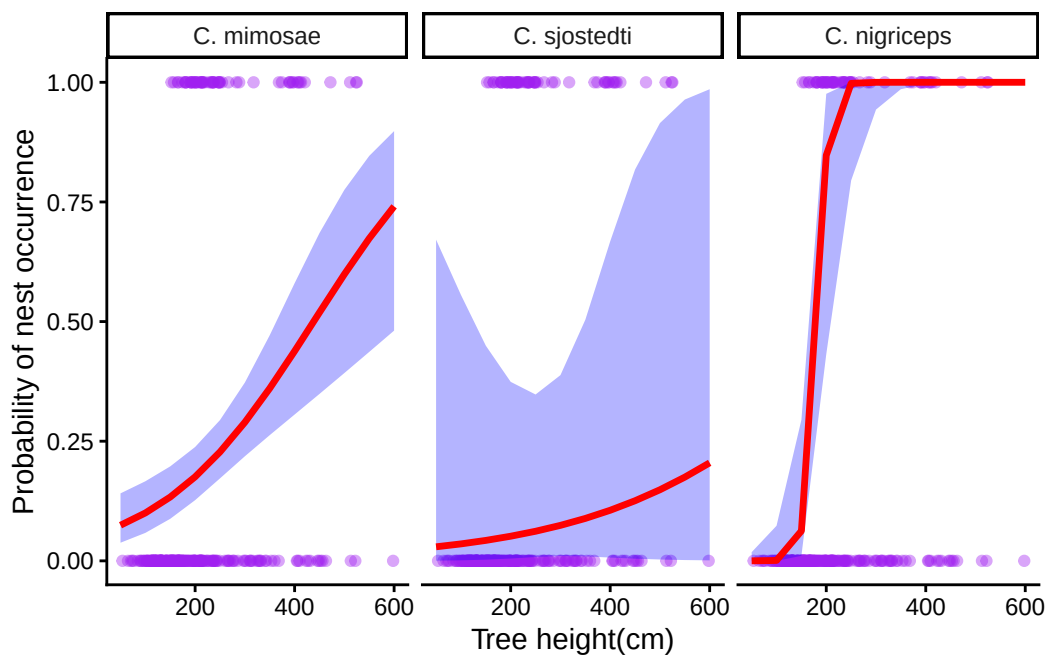
```
# base layer: ggplot
ggplot() +
  # first layer: jitter
  geom_jitter(
    data = nests_clean,
    aes(
      x = height_cm,
      y = case_control
    ),
    # setting customizations
    width = 0,
    height = 0,
    color = "purple",
    alpha = 0.4
  ) +
  # second layer: ribbon
  geom_ribbon(
    data = predictions,
    aes(
      x = x,
      ymin = conf.low, # setting CI
      ymax = conf.high
    ),
    fill = "blue",
    alpha = 0.3
  ) +
  # third layer: line
  geom_line(
    data = predictions,
    aes(
      x = x,
      y = predicted
    ),
    color = "red",
    linewidth = 1.2
  )
```



```

) +
# seperating by ant species
facet_wrap(~group) +
# labeling x and y axis
labs(
  x = "Tree height(cm)",
  y = "Probability of nest occurrence"
) +
# retheming
theme_classic() +
  # removing legend
  theme(
    legend.position = "none"
  )

```



i.

Predicted probability of a bird nest being in a tree depending on the height and species of acacia ant. Purple points represent the data points of whether or not there was a nest in the tree. Red line shows the predicted probability at each tree height. The blue ribbon shows the 95% confidence interval. The first panel shows data for the *C. mimosae* species, the middle for the *C. sjostedti* species, and the third one for *C. nigriceps*. Each shows an increase in probability as

height increases with *C. nigriceps* being the steepest curve, then *C. mimosae*, then *C. sjostedti*. The data is from Palmer et al (2023).

j.

```
# new object
pred_600 <- data.frame(
  height_cm = 600, # setting predictor
  scientific_name = unique(nests_clean$scientific_name)
)
# predicting the probabilities
predict(
  model2,
  newdata = pred_600,
  type = "response")
```

1	2	3
0.7407857	0.2048017	1.0000000

k.

At the height of 600 cm, the predicted probability of the *C. mimosae* species trees having a nest is 0.74, the *C. sjostedti* trees is 0.20, and the *C. nigriceps* is 1.00. This means that the third species trees have the highest probability and the second has the lowest. It seems that the *C. sjostedti* species causes trees to be less likely to have a nest in them and the *C. nigriceps* species is the most likely to have a nest in them. This can be explained by the defensive strategies of the different ant species which is also said in the paper. Different species have different more aggressive ways to defend their trees which would stop the birds from nesting there. We can see that the *C. sjostedti* species must be the most aggressive, having the least likely probability of having a nest.

Problem 3.

a.

In one of my graphs in homework 2 I graphed the time of day versus the commute length but it had very few data points and did not have the same shape as my final visualization so the data appears differently. They all are very similar in what they are presenting, none of them show variables that significantly change the trends shown. I see similar patterns in the overall trends

of the data, with highest times early in the morning and the later times being lower commute lengths. In week 9 I got feedback on how to show the weather on my affective visualization because I was not sure how to convey it. I got advice to make it look cloudy on the painting in the morning and in the evening and did that by creating a gradient in the sky on the final painting.

b.

I want to use a linear regression to see the relationship between time of day and commute length. I could use this because I want to see if there is a linear relationship between leaving later in the day and a shorter drive. The variables are time of day which is a continuous quantitative variable and commute length which is also a quantitative continuous variable.

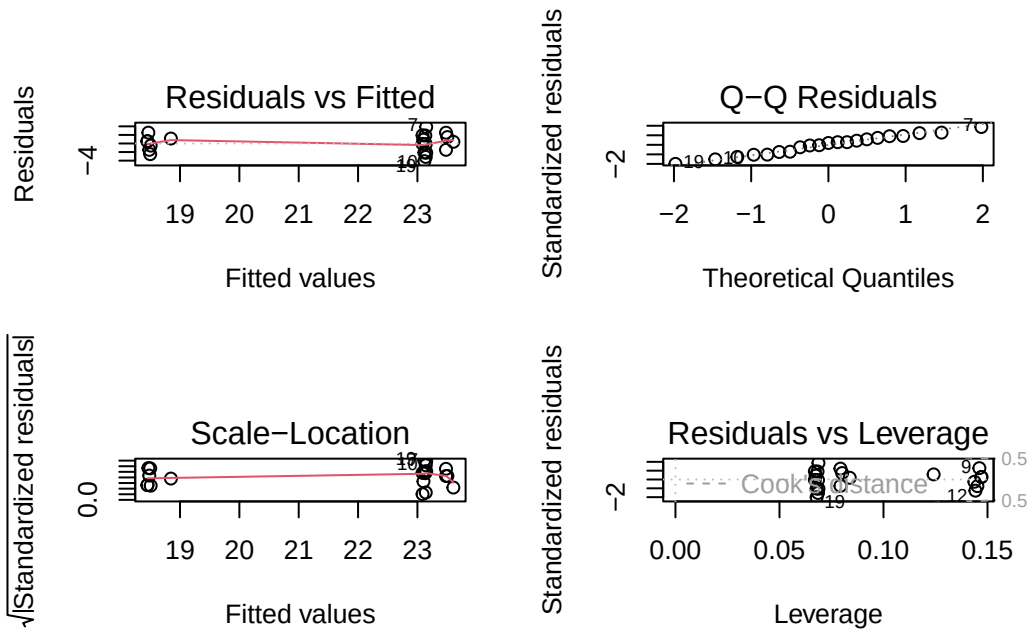
c.

Checking Assumptions:

```
# reading in personal data
personal_data <- read.csv("envs193ds-personal_data - Sheet1 copy.csv")

# creating fit model
data_model <- lm(
  commute.length..min. ~ time.of.day..hr.min.,
  data = personal_data
)

# running diagnostics
par(mfrow = c(2, 2))
plot(data_model)
```



Running Analysis

```
# running test summary for linear model
summary(data_model)
```

Call:

```
lm(formula = commute.length..min. ~ time.of.day..hr.min., data = personal_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.118	-1.482	0.394	1.495	3.851

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.357393	1.677317	17.503	3.54e-13 ***
time.of.day..hr.min.	-0.007751	0.001624	-4.773	0.000132 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.136 on 19 degrees of freedom

Multiple R-squared: 0.5453, Adjusted R-squared: 0.5213

F-statistic: 22.78 on 1 and 19 DF, p-value: 0.0001322

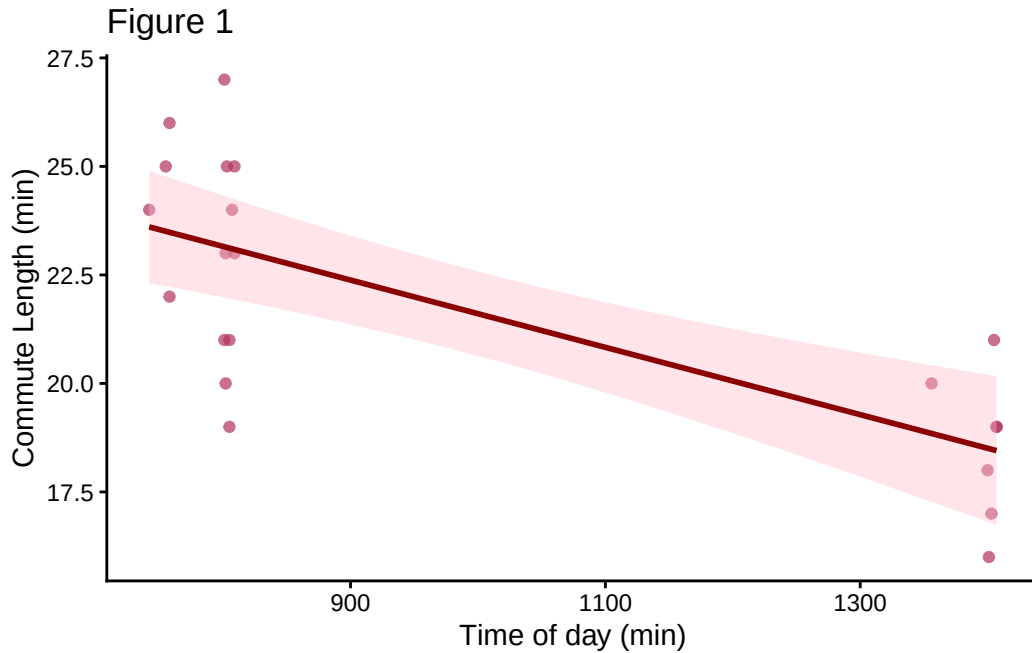
Effect Size

Out of the results from the analysis section I found the R^2 value to be 0.5453 which means there is a positive correlation that is not very strong.

d.

```
# base layer: ggplot
ggplot(personal_data,
      # setting x and y components
      mapping = aes(x = time.of.day..hr.min., y = commute.length..min.)) +
# first layer: geom point
geom_point(color = "maroon", alpha = 0.7) +
# second layer: line and ribbon
geom_smooth(method = "lm", # setting linear
            color = "darkred",
            fill = "pink") +
# labeling axis
labs(x = "Time of day (min)", y = "Commute Length (min)", title = "Figure 1") +
# retheming
theme_classic()
```

`geom\_smooth()` using formula = 'y ~ x'



e.

In Figure 1 the relationship between time of day and commute length is shown in a linear regression model. There are maroon points representing the individual data points. The dark red line shows the linear model and the pink ribbon the 95% confidence interval. The plot shows that as time of day increases, the commute length decreases.

f.

Commute length decreases as time of day increases (linear regression, $\beta = -0.0078$ min/min of day, $t = -4.77$, $p < 0.001$). The later times of day were associated with shorter commutes. The model predicted that the decrease was about 0.47 minutes per hour of the day. The R^2 value of 0.545 shows that there is a moderate connected between variables.

g.

Shown in workshop 10.